

ONLINE COMPLAINT REGISTRATION SYSTEM

Project Documentation

Submitted By

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Full Stack Development (MERN)

Internship Project

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1. Introduction

- **Project Title:** Online Registration Complaint System
- **Team Members:** KACHANA JAYA KEERTHANA (Individual Submission)-Full Stack Developer
- The Online Complaint Registration System is a web-based application developed using the MERN Stack. It allows users to register complaints online through a simple and user-friendly interface. The system helps reduce manual paperwork by maintaining complaint records digitally. Users can securely log in, submit complaints, and manage complaint details efficiently.

2. Project Overview

- **Purpose:** The goal of this project is to provide a platform for users to easily register their complaints online and for administrators to manage those complaints efficiently.
- **Features:** Key functionalities include user login/registration, complaint submission (Create), tracking complaint status (Read/Update), and an administrative dashboard for oversight.

3. Architecture

- **Frontend:** The frontend is built using React.js for a responsive user interface.
- **Backend:** The backend utilizes Node.js and Express.js to manage API endpoints and server logic.
- **Database:** MongoDB is used for data storage and management.

4. Setup Instructions

- **Prerequisites:** Requires Node.js (v18+), MongoDB (Compass/Atlas), and VS Code.
- **Installation:**
 - Clone the repository using `git clone <your-repo-url>`.
 - Install dependencies by running `npm install`.
 - Set up your MongoDB connection string within the `.env` file.

5. Folder Structure

- **Client:** Contains the src folder, which holds components, pages (e.g., Login, Dashboard), and API service calls.
- **Server:** Organized into controllers, models (for database schema), routes (for API paths), and the server.js entry file.

6. Running the Application

- **Frontend:** Navigate to the client directory and run npm start.
- **Backend:** Navigate to the server directory and run npm start.

7. API Documentation

- POST /api/register – Endpoint for new user registration.
- POST /api/login – Endpoint for user authentication.
- POST /api/complaints – Endpoint for submitting a new complaint.
- GET /api/complaints – Endpoint to retrieve a list of all complaints

8. Authentication

- The project uses JSON Web Tokens (JWT) to manage user sessions and ensure secure authorization. Each protected request is verified through the authorization header.

9. User Interface

- **UI Features:** The interface features a clean, responsive design using CSS, ensuring accessibility on both desktop and mobile devices.

Welcome Back

Email Address

keerthana@example.com

Password

Login

Don't have an account? [Register here](#)

User Dashboard

Logout

File a New Complaint

Complaint Title

e.g. Room fan not working

Category

Network Issues

Detailed Description

Describe your issue here...

Submit Complaint

Your Complaints History

No complaints filed yet.

ADMIN Complaint Control Center

Logout

All Registered System Complaints (Admin View)

User Details	Complaint Details	Category	Current Status	Action
Keerthana user@example.com	Hostel fan issue Fan regulator is not working and making noise in room number 102	Electrical	Resolved	Resolved
Keerthana keerthana@example.com	Wi-Fi router malfunction in Block-c interminnet internet access on the second floor.needs technical inspection	Network Issues	Pending	Pending
Keerthana keerthana@example.com	AC is not working AC in room 104 is not cooling. Needs gas refill or service	Electrical	Pending	Pending

10. Testing

- **Strategy:** Manual testing was performed for the complete user journey, including registration, complaint submission, and status updates.

- **Tools:** Postman was utilized to test the functionality and response of the backend API endpoints.

11. Screenshots or Demo

- **Demo:** The live source code and project implementation can be accessed on GitHub at: <https://github.com/Keerthana11309/online-complaint-registration-system>
- **Visuals:** You can find screenshots of the application, including the login flow, complaint submission form, and administrative dashboard, within the repository's documentation folder

12. Known Issues

- **Real-time Updates:** The application does not currently feature a real-time notification system, requiring users to manually refresh their dashboards to view status updates.
- **Session Management:** Developers and users should note that the system requires users to re-login if the current JSON Web Token (JWT) expires.

13. Future Enhancements

- **Real-time Notifications:** Future iterations could integrate Socket.io to provide users with instant alerts regarding their complaint status.
- **User Management:** Planned improvements include adding comprehensive user profile management features.
- **Communication Features:** Future updates may include automated email alerts to notify users whenever their complaint status changes.
- **Admin Dashboard:** The administrative dashboard could be enhanced with advanced search and filtering mechanisms to improve efficiency.

